PROJECT PROPOSAL · MULTI-AGENT AI CUSTOMER SUPPORT

Intelligent Support Engine

One chat window for the customer. Ten specialised agents behind it, orchestrated by LangGraph, resolving refunds, technical questions and complaints in under 90 seconds, and handing angry customers to a human before they churn.

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Duration: 4 weeks
Stack: FastAPI · LangGraph · Groq · Next.js

TEAM Group members

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PROBLEM Support doesn't scale, and bots frustrate

Human-only desks grow cost linearly with ticket volume, and response times spike at peak hours. Rule-based bots ("press 1 for...") deflect customers who actually need help, which raises re-contact rates and lowers CSAT.

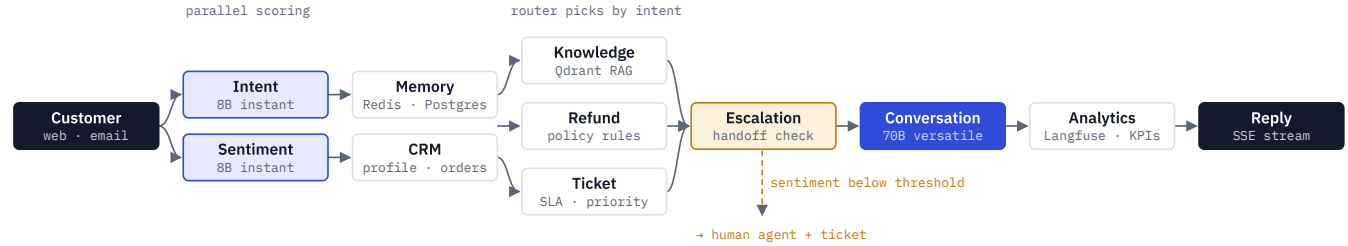
SOLUTION Specialists, not one giant prompt

ISE splits each conversation into focused jobs. Small, fast models classify intent and emotion; a large model writes the reply only after context, policy and actions are settled. **Every decision is traced in Langfuse**, so supervisors can see why the AI refunded, ticketed or escalated. Backend and all 10 agents are already built and passing tests on Groq; this proposal covers the 4 weeks to a working MVP on web chat and email.

MVP TARGETS From the PRD

≥ 60% Containment: resolved with no human	**< 90 s** Median resolution, simple intents
≥ 4.0/5 CSAT on AI-resolved chats	**< 2%** Hallucination on policy answers
< 15% 7-day re-contact, same issue	**99.5%** Availability, single region

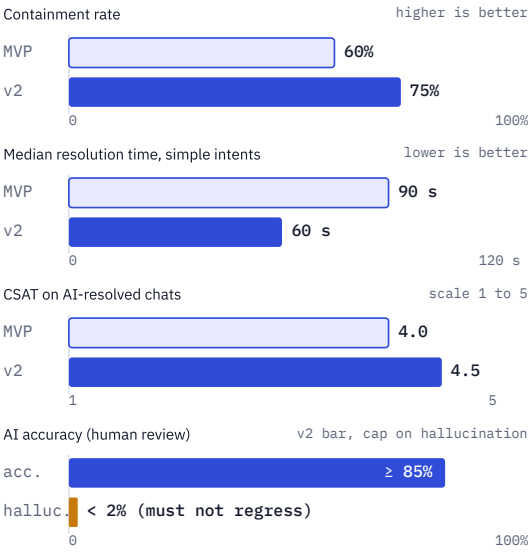
ARCHITECTURE How one message flows through the agents



AGENTS What each agent does, and when it runs

AGENT	JOB	ENGINE	REFUND	TECH Q	BILLING	ANGRY
Intent	Classifies request: refund, technical, billing, complaint	Llama 8B	●	●	●	●
Sentiment	Scores emotion and frustration each turn	Llama 8B	●	●	●	●
Memory	Loads history, open tickets, past orders	Redis/PG	●	●	●	●
CRM	Fetches profile and VIP tier (mocked in MVP)	Mock API	●	●	●	●
Knowledge	Searches the help base, cites sources	Qdrant	○	●	●	○
Refund	Checks eligibility window, issues refund	Rules	●	○	○	○
Ticket	Opens ticket with priority and SLA	Postgres	○	○	●	●
Escalation	Hands off to a human with full context	Threshold	○	○	○	●
Conversation	Writes the final reply in brand voice	Llama 70B	●	●	●	●
Analytics	Logs FRT, resolution, CSAT, traces	Langfuse	●	●	●	●
Agents used per conversation path			**7**	**7**	**8**	**8**

TARGETS MVP now, v2 after launch



AGENT DEEP DIVE

Name, working and techniques of each agent

01 Intent Agent

Llama 3.1 8B · temp 0.1

Reads the customer's latest message and sorts it into one of eight intents: refund, complaint, technical, billing, password reset, subscription, shipping or account deletion. It returns a confidence score that later agents use to decide whether to trust the label.

INPUT	PROCESS	OUTPUT
Latest message	Zero-shot classify, parse JSON	intent, confidence

- Zero-shot prompting
- JSON-constrained output
- Low temperature
- Safe fallback on parse error

02 Sentiment Agent

Llama 3.1 8B · temp 0.1

Scores the emotion of every turn as happy, neutral, frustrated or angry. A frustrated or angry label raises an early escalation flag, so the system reacts to rising anger before the customer leaves.

INPUT	PROCESS	OUTPUT
Latest message	Label-set prompt, extract label	sentiment, escalate flag

- Per-turn scoring
- Fixed label set
- Keyword label extraction
- Threshold flag

03 Memory Agent

Redis · PostgreSQL

Loads what the system already knows about the customer: the last five messages (trimmed to 100 characters each), open tickets and past orders. Redis holds the short-term session; PostgreSQL holds long-term history.

INPUT	PROCESS	OUTPUT
user_id, history	Window last 5, trim, fetch DB	context object

- Sliding-window memory
- Short + long-term store
- Context truncation

04 CRM Agent

Mock API (MVP)

Enriches the conversation with the customer profile: VIP status, membership tier (gold or standard), lifetime value and purchase count. Mocked in the MVP behind an adapter so a real CRM can plug in later.

INPUT	PROCESS	OUTPUT
user_id	Profile lookup, tier rule	crm_data, VIP tier

- Profile enrichment
- Tier-based personalisation
- Adapter pattern

05 Knowledge Agent

Qdrant + Llama 70B

Answers policy and how-to questions from the help base. It embeds the question, retrieves the closest chunks from Qdrant, and asks the large model to answer only from those chunks, citing each source.

INPUT	PROCESS	OUTPUT
Question	Embed, top-k search, grounded answer	answer + sources

- RAG
- Vector similarity search
- Grounded prompting
- Source citation

06 Refund Agent

Deterministic rules

Runs only for refund intents. Checks the order against the refund policy (eligibility window, item condition) and records the refund with amount, reason and approval time. No LLM touches the money path.

INPUT	PROCESS	OUTPUT
intent, order	Policy rule gate	refund status, record

- Rule engine
- Policy gate before payout
- Audit logging

07 Ticket Agent

Rules · PostgreSQL

Opens a ticket (for example TIX-4F2A) when a case needs follow-up. Priority comes from an intent × sentiment matrix: an angry refund is urgent, an angry complaint is high, everything else medium. Category is mapped from intent.

INPUT	PROCESS	OUTPUT
intent, sentiment	Priority matrix, category map	ticket, priority, SLA

- Rule-based triage
- Priority matrix
- Category mapping
- SLA assignment

08 Escalation Agent

Threshold rules

Decides when a human must take over. Three triggers: a frustrated or angry customer, a complaint the Intent agent is unsure about (confidence below 0.6), or an urgent ticket. It hands over a summary of the last three messages so the human starts with context.

INPUT	PROCESS	OUTPUT
sentiment, intent, ticket	3 trigger checks	handoff + summary

- Multi-trigger thresholds
- Confidence gating
- Context summary handoff

09 Conversation Agent

Llama 3.3 70B · temp 0.7

Writes the reply the customer sees. Its system prompt is built at run time from the detected intent, sentiment and context, with the last ten turns of history. Replies stay short (up to three sentences), acknowledge feelings, and stream token by token.

INPUT	PROCESS	OUTPUT
state, last 10 turns	Dynamic system prompt, generate	streamed reply

- Dynamic prompt building
- Tone conditioning
- 10-turn context window
- SSE streaming

10 Analytics Agent

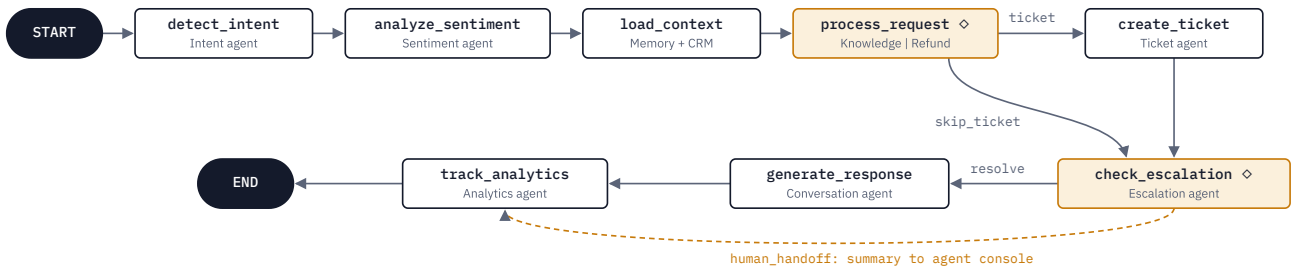
Langfuse · structlog

Runs last on every path. Tags the outcome (resolved, escalated, small talk or needs assistance), counts messages, records refunds, tickets and escalations, and sends the full trace to Langfuse for the supervisor dashboard.

INPUT	PROCESS	OUTPUT
final state	Outcome tag, metrics	KPIs, trace

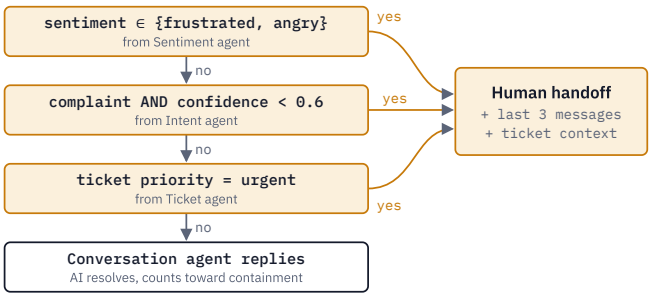
- Structured logging
- LLM tracing
- Outcome tagging
- KPI rollup

ORCHESTRATION LangGraph state machine, as built in `orchestrator/graph.py`



◇ = conditional edge. All agents read and write one shared typed `AgentState` (messages, intent, sentiment, context, ticket, refund, escalation, analytics), so each node sees what earlier nodes decided.

ESCALATION When a human takes over



TRIAGE Ticket priority matrix

	happy	neutral	frustrated	angry
refund	medium	medium	URGENT	URGENT
complaint	medium	medium	high	high
technical	medium	medium	medium	medium
billing	medium	medium	medium	medium

Frustrated / angry also trigger escalation.

MODEL ROUTING Only 4 of 10 agents call an LLM

8B × 2	70B × 2	No LLM × 6

8B: **Intent, Sentiment**
70B: **Knowledge, Conversation**
No LLM: **Memory, CRM, Refund, Ticket, Escalation, Analytics**

LLM calls per message	3 to 4
Classification temperature	0.1 (consistent labels)
Reply temperature	0.7 (natural tone)

Cheap fast model where speed matters, large model only for the words the customer reads, and plain rules wherever money or SLAs are involved.

TECH STACK Layer by layer

CHANNELS	Next.js web chat · Email · WhatsApp (stub)
API	FastAPI · SSE streaming · Clerk auth
ORCHESTRATION	LangGraph StateGraph · Typed AgentState
MODELS	Groq Llama 3.1 8B · Groq Llama 3.3 70B · LiteLLM fallback
DATA	PostgreSQL + Alembic · Redis · Qdrant
OBSERVABILITY	Langfuse · structlog
INFRA	Docker Compose

PLAN Four weeks to MVP

WORKSTREAM	WK 1 · OCT 19	WK 2 · OCT 26	WK 3 · NOV 2	WK 4 · NOV 9
Infra & data	Docker Compose stack, Alembic migrations, seed KB into Qdrant			
Agents & orchestration	Env-based secrets, LiteLLM routing 8B/70B	Refund rules, escalation threshold, ticket SLAs, Langfuse traces		
Frontend & channels		Clerk sign-in, chat UI on SSE API	Email channel, human handoff console, analytics dashboard	
Evaluation			Build 200-conversation test set across 4 paths	Score containment, CSAT, hallucination; tune prompts
Launch				Load test, docs, recorded demo

Milestones: Wk 1 stack boots with one command · Wk 2 refund flow works end to end · Wk 3 angry customer reaches a human · Wk 4 MVP targets measured and demo shipped.

IMPACT UN Sustainable Development Goals

8 — Decent work & economic growth
AI handles repetitive refunds and FAQs so human agents spend their time on complex, higher-value cases, without burnout from queue spikes.

9 — Industry, innovation & infrastructure
Open-source stack and free-tier Groq models make enterprise-grade support affordable for small businesses, not only large firms.

10 — Reduced inequalities
24/7 help in plain language, with Urdu documentation already in the repo, reaches customers outside office hours and outside English.

16 — Accountable institutions
Every AI decision is traced and reviewable in Langfuse, and frustrated customers always reach a human.

Scope: web chat and email. CRM and refund gateway mocked; WhatsApp, Slack and voice are stubs.

Risks: LLM rate limits (fallback via LiteLLM) · wrong refunds (rule gate before any payout)